

Finite Agent Machines

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Abstract

TBD

1 Introduction

Conclusively verifying software outcomes is fundamentally unsolvable for most software. This is evidenced by the sheer number of techniques used to incrementally assure software is correct including unit tests, in-code assertions, human review, compiler and type checking, static analysis and linters, fuzzing, ... etc. Conclusive verification is the bottleneck that prevents brute forcing solutions or otherwise. Large Language Models are improving the ability to search the space of software solutions for improvements in algorithms and software more generally. We say that a solution is *conclusively verifiable* if the checker guarantees that incorrect outcomes are never accepted (soundness), correct outcomes are not rejected (completeness), and the checker always returns an answer for every candidate (totality). For problems where you have a conclusive verifier, the cost of solving that problem boils down to the cost running the verifier and the cost of searching the space of solutions for problems worth checking. If the gains of finding a solution is greater than the estimated cost to search and the cost to verify, then it is worth it to do the search.

In this paper we propose a novel solution to perform solution search when there is no conclusive verifier with cheap cost. Our solution is a novel multi-agent policy with human-oversight.

the main question is whether the benefit of an improvement to a known solution to a problem is worth the cost of searching the space of possible solutions (cite alpha evolve here, and open evolve). For many problems in research and software development, complete verification is unattainable, and this limits the ability to apply LLM-based agents, because every potential solution generated by the agents produces more verification work, and this becomes prohibitively expensive.

There's other issues here that are relevant... while we want agents to be all powerful, the reality is that they still have a number of limitations in practice. Context window sizes, following instructions generally, especially on problems in domains for which they have not been trained well. (this is why benchmarking is a thing). The scope of a problem

given to an agent is incredibly important. From a practitioners point of view, small scoped tasks are much more likely to be executed well by an agent than scope involving larger tasks. This is why the notion of 'sub-agents' have become so popular. Sub-agents are a way to pass out narrow focused tasks so that each agent has a small piece of work to focus on.

Most sub-agents in use today refers to an automatic feature driven by a larger model.

When verification is prohibitively expensive, we need to reduce the number of candidate solutions generated by agents, so that verification costs are within bounds of the budget of an individual or team. On this spectrum, the most extreme version is for all code to be manually reviewed by a human, and on the other end of the spectrum are techniques to limit the space of what agents can do.

We present a novel multi-agent architecture to solve alignment and verification problems with Large Language Models and similar neural-based models.

The ideal use case is work that cannot be trivially or automatically (including with software) verified. Additionally, the consequence of getting the work right is high stakes. For that reason, we avoid the problem of prototyping. We also are interested in work that is not possible to verify the outcome after the fact. Therefore, since the outcome of applying AI to solve a problem is not completely verifiable, the process that produced it offloads some of the trust that a human puts into verifying it. If they can verify the process that was followed by the AI, then they can reduce the amount of verification needed on the end result.

There is a spectrum of approaches to perform concrete work. You could create a single agent, you could use dynamic workflows like Claude Code, you could do everything by hand yourself.

2 Finite Agent Machines

Inspired by finite-state machines (citation), we define Finite Agent Machines. The term *finite* in finite-state machine refers to the finite number of states and transition labels, not to whether execution terminates. Cycles can produce arbitrarily long or infinite runs.

Definition 1. A *finite-state transition system* is a tuple

$$M = (Q, \Sigma, \delta, q_0),$$

where Q is a finite set of states, Σ is a finite set of transition labels,

$$\delta \subseteq Q \times \Sigma \times Q$$

is a transition relation, and $q_0 \in Q$ is the initial state.

A Finite Agent Machine retains this structure but replaces the generic transition-label set Σ with a set $\mathcal{A}$ of agentic transition labels. Each $\alpha \in \mathcal{A}$ identifies a predefined agentic transition.

Definition 2. An *agentic transition* is a tuple

$$\alpha = (\eta, \iota, \mu),$$

where η is an execution harness, ι is an instruction specification, and μ is a model. The harness η constructs the execution context from the current state, provides the permitted tools, and invokes model μ under instructions ι .

Definition 3. A *Finite Agent Machine* is a tuple

$$M = (Q, \mathcal{A}, \delta, q_0),$$

like a finite state machine, except the transitions are agentic.

3 Finite Agent Machines

Inspired by finite state machines (citation) we define finite agent machines here. The 'finite' in finite state machines refers to the fact that there is a finite number of states and a finite number of transitions, not whether a finite state machine terminates. It is easy to construct a finite state machine that can run for a very long time or forever. Likewise, 'finite' in Finite Agent Machines refers to a finite set of transitions between states, where each transition between states is the result of an agent running. While this could be applicable to non-software based work, we describe a state space and agentic transitions for a software development domain.

Definition 4. A *finite-state transition system* is a tuple

$$M = (Q, \Sigma, \delta, q_0),$$

where Q is a finite set of states, Σ is a finite set of transition labels, $\delta \subseteq Q \times \Sigma \times Q$ is a transition relation, and $q_0 \in Q$ is the initial state. Cycles in δ permit arbitrarily long or infinite runs, so finiteness of the system does not imply termination.

A finite agent machine is one where transitions are agentic steps. Unlike other planning formalisms where transition functions may be well defined in terms of preconditions and effects, agentic transitions are more loosely defined since the transition operation is expected to be *intelligent*. Agentic transitions still specify when the agent transition may run, which inputs it receives, which outputs may be accepted, and how an accepted output changes the machine state.

Definition 5. Let Q be a set of states and let Σ be a set of transition labels. For $\alpha \in \Sigma$, an *agentic transition*

$$q \xrightarrow{\alpha} q'$$

is a transition in which the Step identified by α invokes an agent from state q , obtains a candidate result, and incorporates an accepted result into successor state q' .

Definition 6 (Agentic transition schema). An *agentic transition schema* is a tuple

$$\alpha = (G_\alpha, E_\alpha, V_\alpha, U_\alpha),$$

where:

- $G_\alpha(\gamma) \subseteq \mathcal{B}_\alpha$ is the set of input bindings for which α is eligible in configuration γ ;
- $E_\alpha \subseteq \mathcal{B}_\alpha \times \mathcal{O}_\alpha$ is the agent-execution relation, where $(b, o) \in E_\alpha$ means that executing the agent on binding b may produce proposal o ;
- $V_\alpha(\gamma, b, o)$ determines whether proposal o is accepted; and
- $U_\alpha(\gamma, b, o)$ constructs the successor configuration obtained by publishing an accepted proposal.

The schema induces a concrete agentic transition

$$\gamma \xrightarrow{\alpha, b, o} \gamma'$$

exactly when

$$b \in G_\alpha(\gamma), \quad (b, o) \in E_\alpha, \quad V_\alpha(\gamma, b, o) = \text{accept}, \quad \gamma' = U_\alpha(\gamma,$$

The relation E_α may be nondeterministic: repeated executions with the same binding may produce different proposals. An execution that fails, does not terminate, or produces a rejected proposal does not induce an agentic transition.

and

Definition 7. A *finite-state agentic-transition system* is a tuple

$$M = (Q, \alpha, \delta, q_0),$$

where Q is a finite set of states, Σ is a finite set of transition labels, $\delta \subseteq Q \times \Sigma \times Q$ is a transition relation, and $q_0 \in Q$ is the initial state. Cycles in δ permit arbitrarily long or infinite runs, so finiteness of the system does not imply termination.

4 One Playbook, Increasing Structure

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5 Execution Model

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6 The Example as a Concrete Trace

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7 Related Work and Positioning

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8 Limitations and Open Decisions

TBD

9 Conclusion

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Appendix: Schematic Playbook Definition

TBD

Previous Draft (Preserved)

Abstract

Long-running agent work produces both a result and an execution process: choices about decomposition, evidence, parallelism, combination, and when no further work is enabled. Synthesizing that process anew makes its decomposition and coordination choices another task-specific output. For recurring work whose results are difficult to check directly, a versioned process definition and an exact execution record give reviewers stable objects to inspect. The technical challenge is to retain reusable structure without forcing every execution down a fixed path.

We present *Reactive Playbooks*, a material-state execution model that separates revision-stable agent task templates, called Steps, from their concrete applications determined at runtime. Exact artifact occurrences and immutable collection members determine which Steps become eligible. Each execution works in isolation and may publish one validated successor state; parallel executions publish siblings, coverage-checked merges record all supplying parents, and follow-up publications reuse Step definitions while extending a forward-only history.

A research-to-design Playbook demonstrates runtime-determined parallel appraisal, question-level synthesis, explicit design merge, and recurrence. An event-driven reference algorithm yields obligations for atomic visibility, completion-order-independent lineage, at-most-once automatic claims, coverage-preserving merge, and forward recurrence; it also defines quiescence as the absence of eligible, queued, or running work. Together, the model and obligations let one versioned definition generate materially different executions while preserving an exact account of which activities ran, the material each activity consumed, and how each result became visible.

1 Introduction

An AI system asked to investigate a production failure, conduct research, or design software must do more than produce a final artifact. The system must instantiate an execution process: decompose the work, decide which evidence to seek, determine what can run in parallel, combine intermediate results, and decide when no further work is enabled. When the final artifact is difficult to evaluate directly, a reviewer also needs to understand that execution process. An execution record cannot prove the artifact correct, but it can expose which activities ran, what material each activity consumed, and how each result became visible.

The execution process may be synthesized for one task or deliberately retained as a versioned definition for repeated use. Task-specific generation avoids the cost of maintaining a reusable procedure when work is novel or one-off. This paper studies the complementary case in which recurring process decisions remain explicit across executions. The distinction concerns lifecycle rather than authorship: either a person or an agent may draft the process. A reusable definition also need not impose a fixed trace; the retained Step definitions can remain stable within a revision while runtime-produced material determines which concrete activities run, fan out, merge, recur, or leave no work enabled.

Retaining the process definition exposes an execution-semantics problem. Agent frameworks can compose models, tools, and humans into useful interaction patterns (Wu et al. 2024), while long-running repository work may require one activity to edit for minutes, exhaust a model context, continue in another model-harness invocation (a *Session*), or finish concurrently with another activity. A saved diagram or set of prompts alone does not determine when intermediate edits become shared, how concurrent results relate, which evidence authorizes a merge, why an earlier Step becomes eligible again, or how reevaluation avoids creating the same application twice. Ambiguous publication, lineage, recurrence, and claim semantics become acute when each activity is a fallible, long-lived agent operating on material files rather than a deterministic function over a small symbolic state.

To make the execution problem concrete, we use a research-to-design Playbook whose purpose is to turn an initial research agenda into an evidence-linked design recommendation. Frame derives decision-relevant questions; Discover identifies candidate sources; Appraise evaluates each question-source pair; Synthesize combines the appraisals for one question; and Design integrates the question-level findings. Discovery may create any finite number of appraisal requests, and Design may publish a targeted follow-up question when another research pass could change the design. The example therefore combines a runtime-determined number of branches, parallel work, explicit merges, and recurrence.

Reactive Playbooks address this execution-semantics problem with a material-state semantics. A Playbook revision contains a validated set of Step definitions. *Grounding* instantiates one Step definition only after exact artifact occurrences—particular publications, even when their bytes match earlier versions—or immutable collection members resolve its input declaration. Before dispatch, a durable claim reserves the definition revision, exact binding, and runtime-policy scope so that the same automatic application is not created twice. The resulting Step execution may span sequential Sessions, but its edits remain private until an explicit completion request passes validation and atomically publishes one immutable state. Every publication wakes evaluation over the recorded history; no active cursor, routing token, or loop counter determines what runs next. One Playbook revision can therefore produce materially different executions while retaining exact material lineage for each execution.

We make three contributions. First, we give a compact execution model that separates readiness evidence, bound inputs, candidate states, selected parents, and recorded lineage. Second, we use one evolving research-to-design Playbook to show how material dependencies produce linear order, late-grounded parallelism, explicit merging, and recurrence while concrete history remains a directed acyclic graph (DAG). Third, we define an event-driven grounding, claiming, and publication algorithm and derive implementation-neutral obligations for a conforming runtime. Together, the model, example, and reference algorithm provide an implementation-neutral foundation for execut-

ing versioned definitions of agent processes across parallelism, merging, and recurrence while preserving exact material provenance.

2 Planning Policies and Persistent Execution

In nondeterministic planning, an action may have more than one possible result, and a policy associates encountered states with actions so execution can respond to the result that actually occurs (Kuter and Nau 2004). Reactive Playbooks are policy-like only in this persistent, reactive sense. A Playbook revision fixes Step schemas—their instructions, runtime configuration, input and readiness declarations, output contracts, and collection relationships—while execution grounds concrete applications from immutable material history. Unlike a conventional planning policy, evaluation may enable several Step applications concurrently, consults provenance and durable claims rather than one active state, and has no goal test or optimization objective. This planning-policy analogy is distinct from the runtime policies for capacity, retries, authority, and storage discussed later.

The execution horizon—the number of concrete Step executions that unfold—is not fixed. Continued publication of fresh bindings can produce arbitrarily long and potentially nonterminating forward histories even though a Playbook revision contains finitely many Step definitions. Planning distinguishes an intended goal state from a *dead end*, a non-goal state from which the goal is unreachable. The Playbook model defines neither category because it has no single active state and no goal predicate. Its closest stopping condition is history-wide quiescence: no binding is eligible and no claimed work is queued or running. Quiescence may coincide with an application’s intended terminal result or with an unproductive stopping point, but the runtime does not decide which; it remains attached and may wake after a later publication or manual event.

3 One Playbook, Increasing Structure

Throughout the paper, the running Playbook begins with a research agenda and produces an evidence-linked design recommendation. Frame turns the agenda or a follow-up handoff into questions; Discover selects plausible question-source pairs at runtime; Appraise evaluates those pairs independently; Synthesize combines the declared appraisals for each question; and Design merges the question-level conclusions. If Design identifies a material unresolved question, it publishes a follow-up handoff and the same five Step definitions apply again; otherwise the Playbook becomes quiescent.

Figure 1 gives a structural overview of this Playbook at three resolutions; Listing 1 in the appendix records its compact schematic definition. Panel (a) introduces the linear material dependencies; panel (b) exposes runtime-grounded Appraise siblings and their explicit Synthesize merge; and panel (c) adds the recurrence arc created when Design publishes a follow-up that grounds another Frame execution. Solid arrows denote material dependencies rather than authored routing. The dashed arc denotes reuse of the same

Step definitions over new bindings, not a backward edge in recorded state history.

A linear first reading

At its coarsest resolution, the running Playbook forms a material-dependency chain: Frame produces questions, Discover produces a source map, Appraise produces evidence, Synthesize interprets the evidence, and Design applies the interpretation. Ordering is a consequence of material availability. Synthesize does not follow Appraise because an edge moved a token; Synthesize becomes groundable because exact appraisal-result occurrences now fulfill its declared input set. The next subsection refines Appraise and Synthesize into collection-grounded fan-out and merge.

The linear reading gives two basic requirements. Intermediate writes must not make downstream Steps applicable, because an agent can revise or abandon them. Conversely, accepted work must have an exact identity, because a filename such as `design.md` may be republished with identical or different bytes during a later pass. The model therefore distinguishes an artifact *occurrence*, identified by its creating execution and state, from its content digest.

Dynamic branching and explicit merging

Discover cannot know the useful fan-out before it inspects the questions and search space. At accepted completion, Discover freezes a finite `appraisal-requests` collection. Each member binds one Appraise execution, so four members yield four independent executions without four authored copies of the Step. The scheduler may run these executions concurrently, subject to a capacity policy that does not change their logical bindings.

A question-level Synthesize execution requires the successful results that fulfill its exact subset of appraisal requests. The present model also requires completion of the entire Discover expansion as a conservative readiness barrier, without treating unrelated appraisal states as inputs or parents. At completion, Synthesize selects a duplicate-free set of authorized states whose result occurrences collectively cover the bound requests. The engine, not the agent, verifies coverage and records those states as parents.

Repetition without cyclic history

Design emits a finite `followup-research` collection. A nonempty collection creates a new exact binding for Frame and therefore another forward research pass. An explicitly empty collection creates no Frame binding. Once no other binding is eligible or running, the Playbook is *quiescent*. Quiescence means that no configured work is currently available; it is not a proof that the design is correct, a goal has been achieved, or later human work cannot wake the Playbook. If every Design execution instead publishes a fresh follow-up member, forward recurrence can continue without a fixed bound even though the recorded history remains acyclic.

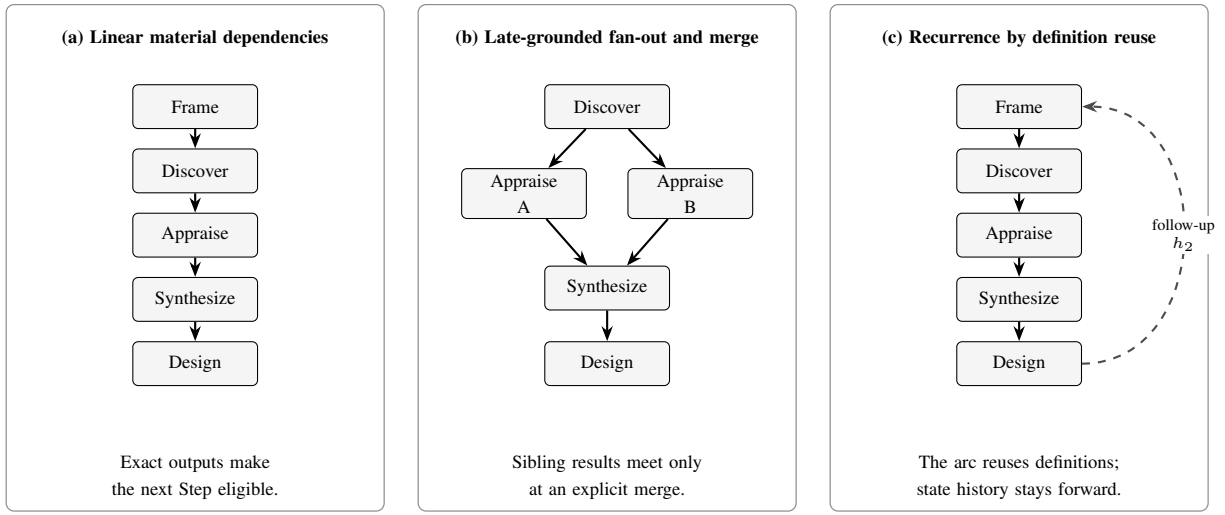


Figure 1: Three structural views of the running research-to-design Playbook. Panel (a) shows the linear dependency chain created by exact outputs. Panel (b) shows Discover late-grounding independent Appraise executions and Synthesize merging their complete result set. Panel (c) adds a dashed loop arc from Design to Frame: follow-up h_2 makes the same Frame definition applicable to a new binding. The arc depicts recurrence in the Playbook definition; every accepted execution still adds a new state along forward-only recorded history.

4 Execution Model

Immutable task states

Let $\mathcal{N}$ be exact artifact names, $\mathcal{O}$ immutable artifact occurrences, and $\mathcal{W}$ material workspace checkpoints. A sealed task state is

$$s = (id_s, w_s, A_s, C_s, Q_s, \rho_s), \quad (1)$$

where id_s is a fresh state identifier and $w_s \in \mathcal{W}$ is a complete checkpoint. $A_s : \mathcal{N} \rightarrow \mathcal{O}$ is a partial map from artifact names to visible occurrences, so a name may be absent; C_s maps collection names to finite frozen member sets; Q_s is a duplicate-free set of already sealed parent-state identifiers; and ρ_s records the producing execution, definition revisions, and bound-input provenance. A state is immutable after publication. Several states may coexist; lineage does not mark one state as the active truth.

An occurrence identifies a publication event, not only bytes. Republishing unchanged bytes can create a new occurrence, while an inherited artifact keeps its prior occurrence identity. Occurrence identity lets a binding state exactly which work it consumed without making every descendant appear fresh.

The state history $H = (S, E)$ contains edge (p, s) for every $p \in Q_s$. Because publication may name only already sealed parents and a new state receives a fresh identity, H is acyclic by construction. An ordinary execution has one mechanically fixed parent. A merge-capable execution may select one authorized parent when that state already covers every bound input; selecting several distinct authorized parents creates a merge transition when the inputs live on siblings.

Definitions, bindings, and collections

A Playbook revision P^r is a finite set of validated Step-definition revisions. A definition d^r contains agent instructions, runtime configuration, input declarations, readiness conditions, an output contract, and collection relationships. Validation rejects unresolved references and incompatible collection keys before any Session starts.

A binding b contains exact ordinary occurrences and collection-member identities. Three grounding forms are sufficient for the running example:

1. an ordinary input resolves an exact named occurrence in one sealed state;
2. `each(C)` enumerates one candidate grounding for every exact member of frozen collection C ; and
3. `complete-set(R)` resolves the exact successful results whose immutable fulfillment records cover required member set R .

Collection membership is never inferred from filename patterns, directory scans, or natural-language parsing. The producer explicitly declares a finite collection during completion; the engine validates and freezes the declaration inside the successor state.

Accepted publication records each declared fulfillment as an exact relation from a result occurrence to the request member it fulfills. Failed or rejected executions record no successful fulfillment. Claims, rather than grounding, suppress duplicate automatic executions for already claimed exact bindings.

Readiness and input resolution are separate operations. Let u be a readiness witness set and b the exact input binding. A broad condition may establish that all Appraise requests from one Discover expansion have some successful result, while a Synthesize binding contains only the results

for one question. The witness states in u do not become input, merge candidates, or lineage unless their occurrences also appear in b .

Automatic duplicate suppression uses a durable claim identity

$$\kappa = \text{ClaimIdentity}(P^r, d^r, b, \chi), \quad (2)$$

where χ is the still-open policy scope, such as ordinary-parent context or an ancestry scope. Its canonical storage encoding is an implementation choice. Claim acquisition and queued-execution creation form one atomic operation. A claim names logical identity; it is not a proof of successful completion. Explicit manual reruns receive distinct execution identities. Failure retention and automatic retry policy remain separate decisions rather than hidden parts of the binding.

Executions and Sessions

A Step execution

$$e = (P^r, d^r, b, A_e, K_e, o_e, W_e, \langle q_1, \dots, q_k \rangle) \quad (3)$$

binds one definition revision to one exact input, a frozen authorized candidate set A_e , required coverage K_e , a mechanically fixed ordinary parent o_e when applicable, an isolated workspace W_e , and an ordered sequence of harness Sessions. At most one Session is write-capable at a time. A continuation Session receives the same definition, binding, and unpublished workspace; it does not acquire another claim or create an intermediate state.

Process lifetime is therefore not the semantic boundary. A Session may exit without completing the execution, and an execution may continue in another Session. Conversely, a still-live process cannot mutate a state after accepted completion. The Step execution, rather than its operating-system process, owns the single publication decision.

Validated atomic publication

An agent requests completion by declaring outputs, collections, and, for a merge, selected parent identifiers. The engine serializes completion against continuation, suspends writes, and checks four conditions:

1. every required output exists at an allowed path;
2. every required collection is present, including an explicitly empty collection where emptiness is meaningful;
3. selected parents are duplicate-free members of the candidate set frozen in e ; and
4. selected parents collectively cover every exact bound input that requires cross-state merge.

After validation, the engine checkpoints the complete workspace, assigns occurrence identities only to declared publications, freezes collections and fulfillment relationships, constructs the state manifest, and atomically makes the state and successful execution visible. A rejected completion publishes nothing and leaves the execution continuable. Explicit failure closes the execution without a successor and records failure outside the successful state DAG.

Algorithm 1: Reactive grounding, claiming, and publication

Require: Playbook revision P^r , sealed history H

```

1: evaluate once when  $P^r$  is attached
2: loop
3:    $I \leftarrow \text{Snapshot}(H)$ 
4:   for all  $d^r \in P^r$  do
5:     for all  $g \in \text{Ground}(d^r, I)$  do
6:        $r \leftarrow \text{ResolveExactInputs}(d^r, g, I)$ 
7:       if  $r = (b, A, K, o)$  then
8:          $(\text{ready}, u) \leftarrow \text{Ready}(d^r, g, b, I)$ 
9:         if  $\text{ready}$  then
10:           $e \leftarrow \text{ClaimAndCreate}(P^r, d^r, r)$ 
11:          if  $e \neq \text{none}$  then
12:            enqueue  $e$ 
13:          end if
14:        end if
15:      end if
16:    end for
17:  end for
18:  dispatch queued executions within capacity
19:  if no unclaimed binding is eligible and no work is
    queued or running then
20:    report quiescence; continue awaiting events
21:  end if
22:   $x \leftarrow \text{AwaitEvent}()$ 
23:  if  $x = \text{Completion}(e)$  then
24:    if  $\text{ValidateAndPublish}(e, x)$  then
25:      add the sealed successor to  $H$ 
26:    else
27:      keep  $e$  continuable with validation errors
28:    end if
29:  end if
30: end loop

```

The publication rules give the central visibility invariant:

$$\begin{aligned} \text{accepted}(e) &\implies |\text{Successors}(e)| = 1, \\ \neg \text{accepted}(e) &\implies |\text{Successors}(e)| = 0. \end{aligned} \quad (4)$$

No downstream evaluation observes a partially checkpointed state or a successful provenance record without its corresponding state.

Reactive evaluation

Algorithm 1 gives the event-driven reference behavior. Grounding and readiness inspect a consistent snapshot of sealed history. Exact input resolution either waits or returns (b, A, K, o) ; the atomic claim-and-create operation freezes those values in the execution record. Claiming occurs before capacity-limited dispatch, so two evaluator passes cannot create duplicate automatic executions for the same identity. A scheduler chooses when queued work starts but cannot change what the binding means.

The loop in Algorithm 1 represents persistent attachment, not busy polling. Implementations reevaluate on state-sealed and explicit manual events, and otherwise block. A state publication may ground many executions; capacity limits dispatch but does not defer claim identity to a mutable future

snapshot. Quiescence is reported after a complete evaluation pass finds no eligible unclaimed grounding and no queued or running execution, regardless of whether the preceding event published a state.

5 The Example as a Concrete Trace

Linear publication boundaries

Let the seed S_0 contain one initial research handoff. Frame executes in an isolated workspace and publishes S_1 with a `research-questions` occurrence. Discover binds that exact occurrence, freezes its source decisions, and publishes S_2 . Until each completion is accepted, the next Step sees neither a half-written question list nor a temporary source map.

The linear trace demonstrates why a full state, rather than an artifact bundle alone, is the unit of publication. An agent may change code, supporting files, and several handoffs together. A complete checkpoint makes the transition auditable and preserves the exact material context for later inspection or a replay attempt, while the artifact map supplies stable logical names.

Fan-out, siblings, and coverage

Here, r_{iX} denotes the appraisal request pairing question i with source X , and S_{iX} is the sibling state publishing the corresponding appraisal. Suppose Discover freezes appraisal requests $r_{1A}, r_{1B}, r_{2B}, r_{2C}$. Four Appraise executions share parent S_2 but receive distinct member bindings. Accepted results create siblings $S_{1A}, S_{1B}, S_{2B}, S_{2C}$ even if their wall-clock completion order differs. The runtime never treats a later-finishing sibling as a descendant of an earlier-finishing sibling.

Synthesize for question 1 binds only successful results fulfilling $\{r_{1A}, r_{1B}\}$. The execution's authorized candidates are frozen when the binding is resolved. At completion, the agent may select the states that actually supply those results; the engine accepts only a duplicate-free set whose union covers both requests. Results for question 2 can serve as a global readiness witness but cannot silently become parents of question 1. The same rule lets Design merge only the exact question syntheses it uses.

Separating readiness, binding, and lineage prevents three plausible errors. First, a state that merely existed when a Session started does not become a parent. Second, a state used to establish global readiness does not become an input. Third, repeating one parent identifier does not satisfy a two-result merge. Parentage records material dependence rather than temporal proximity.

Another pass

After Design publishes S_D , a follow-up member h_2 names the exact research question that could materially change the design. The member is a new binding, so Frame can execute again under the same definition revision. Subsequent Discover and Appraise executions create new descendants with new occurrence and collection identities, even if a question or source key reuses the same human-readable text.

If Design instead freezes `followup-research` as the empty set, no Frame execution is grounded. The Playbook becomes quiescent after all other claimed work closes. A future manual-publication operation could add another hand-off and wake the same Playbook without rewinding or deleting earlier history; its authority and interface remain open.

Semantic obligations exposed by the trace

The example yields five obligations for a compatible runtime:

1. **Atomic visibility:** each accepted execution publishes one full state; rejected or failed work publishes none.
2. **Topology independence:** scheduler order changes latency, not sibling or parent relationships.
3. **Automatic-claim uniqueness:** each automatic claim identity κ is acquired at most once.
4. **Coverage-preserving merge:** selected parents are authorized and collectively supply the bound result occurrences.
5. **Forward recurrence:** repeating a Step definition creates a new descendant; no concrete history edge points backward.

Any runtime claiming compatibility with the model must preserve these obligations. The obligations concern execution provenance and publication topology; the semantic quality of an appraisal or design remains outside the execution specification.

6 Related Work and Positioning

Classical planning supplies precise vocabulary for states, action applicability, and successor transitions (Fikes and Nilsson 1971). Hierarchical task network planning supplies authored task decompositions that a planner refines toward executable networks (Erol, Hendler, and Nau 1994). Reactive Playbooks use neither goal search nor search over alternative Step schemas or task networks: the process definition is fixed, while agent-produced data determines how many fixed Step executions are late-grounded. The model is therefore closer to execution and monitoring of an authored process than to plan synthesis.

Workflow languages commonly represent splits, joins, choices, and loops as control-flow patterns (van der Aalst et al. 2003); statecharts likewise make control state and transitions explicit (Harel 1987). A Playbook may be rendered with similar shapes, but its source of truth is exact material dependency and provenance. The definition view may contain a visual cycle, whereas the execution view contains only newly published DAG nodes. Avoiding a cursor is not a claim that workflow formalisms are incapable of the example. Material-state sufficiency is a design choice that makes it possible to reconstruct why each agent execution occurred.

Temporal planning for actions with duration distinguishes an action's interval of execution from the points at which its effects change the state (Fox and Long 2003). A Playbook Step execution draws a narrower analogous boundary: several sequential harness Sessions may perform work, but accepted completion exposes one atomic successor. The paper

does not adopt the temporal constraints, time-varying numeric state variables, or formal plan validation of the Planning Domain Definition Language (PDDL).

Multi-agent frameworks such as AutoGen provide flexible conversation and tool interaction patterns (Wu et al. 2024). A conversational agent can run inside a Playbook Session; the Playbook model instead governs durable publication between executions. General provenance models describe entities, activities, agents, and derivations (Moreau and Missier 2013). Playbook provenance specializes that vocabulary with exact input bindings, fulfillment relationships, automatic claim identity, and coverage-checked state parents.

7 Limitations and Open Decisions

This paper specifies execution semantics and illustrates their consequences with a constructed trace. The example demonstrates that the model represents this research-to-design pattern; it does not establish a general expressiveness result, scalability, liveness under failure, agent-output quality, implementation correctness, or operational security. Claims about those properties require separate implementation and evaluation work.

Several mechanisms remain policy rather than semantics. A production specification must settle claim scope across sibling ancestry, retention and retry after failure, deterministic handling of multiple valid fulfillments, empty complete-set behavior, merge workspace conflicts, scheduler fairness, and the representation of manual state creation. Durable publication also requires a concrete transaction and recovery protocol that prevents a checkpoint, manifest, claim, and database record from disagreeing after a crash. External side effects cannot be rolled back merely because filesystem publication fails.

The model currently assumes one authority creates automatic claims for a task. Multiple evaluators would require fencing or a shared transactional store. An isolated workspace is a Playbook visibility boundary, not a security sandbox; network, subprocess, and external-tool confinement remain outside the present semantics. Human interaction is allowed inside a Session and through explicit manual continuations, yet the paper does not define approval policies or how a human selects a historical state in the interface.

Finally, the schematic notation in Appendix A is not a frozen serialization format. The semantic distinctions are the intended contract: finite explicit collections, exact bindings, atomic claims, isolated executions, validated completion, and coverage-preserving lineage. Locking a portable syntax before the recovery and authority decisions would turn unresolved policy into accidental compatibility requirements.

8 Conclusion

Reactive Playbooks model long-running agent work as provenance-bound transitions over immutable material states. One research-to-design definition consequently expresses linear handoffs, runtime fan-out, concurrent siblings,

coverage-preserving merge, and later refinement without a control cursor or cyclic history. The reference algorithm makes the governing obligations explicit: exact grounding, atomic claiming, isolated execution, validated single-state publication, and fail-closed parent coverage. Recovery, authority, resource, retry, and serialization policies remain open, but implementations can resolve those policies against the stable semantic boundary defined here.

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A Schematic Playbook Definition

Listing 1 records a compact definition of the running example. The notation names the semantic relationships used in the paper without freezing a portable serialization format. Agent instructions, harness selection, resource policy, and output paths remain part of a complete Step definition even where the schematic elides them.

Listing 1: Schematic research-to-design Playbook. An initial agenda is represented as the first member of `followup-research`; later Design executions may emit another member or an explicitly empty collection.

```
playbook: research-to-design
collections:
  appraisal-requests: key(question, source)
  appraisal-results: fulfills(appraisal-requests)
  question-synthesis-requests: key(question)
  question-syntheses: fulfills(question-synthesis-requests)
  followup-research: key(handoff)
  framed-questions: fulfills(followup-research)
  design-requests: key(design)

steps:
  frame:
    input: each(followup-research)
    output: framed-questions
  discover:
    input: each(framed-questions)
    emit: [appraisal-requests,
           question-synthesis-requests,
           design-requests]
  appraise:
    input: each(appraisal-requests)
    output: appraisal-results
  synthesize:
    input: each(question-synthesis-requests)
    complete-set: required-appraisals(input)
    when: all-appraisals-from-expansion-fulfilled
    output: question-syntheses
  design:
    input: each(design-requests)
    complete-set: required-syntheses(input)
    output: [design.md, followup-research]
```
